

SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME: DATA ANALYSIS WITH SPSS/SAS/R

PRACTICAL 9

AIM: Performing text manipulation using `str_sub()`, `str_split()` (R). import dataset.

>COMMANDS USED

```
library(stringr)
```

```
library(tidyr)
```

```
library(dplyr)
```

```
#
```

```
-----
```

```
-----
```

```
# 1. IMPORT AND SAMPLE DATASET
```

```
#
```

```
-----
```

```
-----
```

```
# Import the CSV file
```

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```
ai_jobs_df <-  
read.csv("AI_Impact_on_Jobs_2030.csv")  
  
# Select a small sample of unique jobs for cleaner  
demonstration  
text_data <- ai_jobs_df %>%  
  distinct(Job_Title, .keep_all = TRUE) %>%  
  slice(1:5) %>%  
  select(Job_Title, Education_Level, Average_Salary)  
  
print("--- Original Sample Dataset ---")  
print(text_data)  
  
#  
-----  
-----  
# 2. USING str_sub() (Substring)  
#  
-----  
-----
```

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Scenario: Extract the first three characters of the job title and the degree level.

Example A: Extract the first 3 characters of the Job Title as a prefix

Syntax: str_sub(string, start, end)
text_data\$Job_Prefix <- str_sub(text_data\$Job_Title, 1, 3)

Example B: Extract the last 6 characters of the Education_Level (using negative indices)

text_data\$Degree_Suffix <-
str_sub(text_data\$Education_Level, -6, -1)

print("--- Data after str_sub() (Prefix and Suffix extracted) ---")

print(text_data %>% select(Job_Title, Job_Prefix, Education_Level, Degree_Suffix))

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#

3. USING separate() (Split String)

#

The `separate()` function automates the splitting of a column into multiple new columns.

We split 'Job_Title' by the space (" ").

```
tidy_data <- text_data %>%  
  separate(Job_Title, into = c("Role_Part1",  
"Role_Part2"), sep = " ", extra = "merge", fill = "right")
```

```
print("--- Data after 'separate' (Job_Title split by  
space) ---")
```

```
print(tidy_data %>% select(Role_Part1, Role_Part2,  
Education_Level))
```

```
library(stringr)
```

```
library(dplyr)
```

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```
library(tidyr)
```

```
#
```

```
-----
```

```
-----
```

```
# 1. RE-LOAD AND SAMPLE DATASET
```

```
#
```

```
-----
```

```
-----
```

```
# The code below re-loads the data and creates the  
same sample used previously
```

```
# for a self-contained demonstration.
```

```
ai_jobs_df <-  
read.csv("AI_Impact_on_Jobs_2030.csv")
```

```
text_data <- ai_jobs_df %>%  
  distinct(Job_Title, .keep_all = TRUE) %>%  
  slice(1:5) %>%  
  select(Job_Title, Education_Level)
```

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```
print("--- Data used for splitting demonstration (First  
3 rows) ---")
```

```
print(head(text_data, 3))
```

```
print("-----")
```

```
#
```

```
=====
```

```
=====
```

```
# 2. USING str_split() (Split String) - The Missing  
Demonstrations
```

```
#
```

```
=====
```

```
=====
```

```
# Scenario: Split 'Job_Title' by the space (" ") into  
multiple components.
```

```
# Method A: Basic Split (Result is a list)
```

```
# This is the raw output, where each split string  
becomes an element in a list.
```

```
split_list <- str_split(text_data$Job_Title, " ")
```

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```
print("--- Method A: Basic Split Output (List format)
---")
```

```
print(split_list[[1]]) # Viewing the split parts of the
first job: "Security Guard"
```

```
print(split_list[[2]]) # Viewing the split parts of the
second job: "Research Scientist"
```

```
# Method B: Split Fixed (Returns a matrix)
```

```
# Using simplify = TRUE converts the output to a
matrix, which is easy to assign
```

```
# back to new columns using standard R subsetting.
```

```
split_matrix <- str_split(text_data$Job_Title, " ",
simplify = TRUE)
```

```
# Assign the split parts to new columns
```

```
manual_split_df <- data.frame(text_data$Job_Title)
names(manual_split_df) <- "Job_Title"
```

```
manual_split_df$Primary_Role <- split_matrix[, 1] #
First column of matrix
```

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```
manual_split_df$Qualifier <- split_matrix[, 2] #  
Second column of matrix (or empty string if single  
word title)
```

```
print("--- Method B: Data after str_split() (Matrix  
Assignment) ---")  
print(manual_split_df)
```

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The screenshot shows the RStudio environment with the following components:

- Environment:** Lists objects in the workspace: `final_list` (5 obs. of 3 variables), `job_title` (3000 obs. of 18 variables), `manual_split_df` (5 obs. of 3 variables), `merged_data` (5 obs. of 7 variables), `range_cols` (3000 obs. of 3 variables), `selected_cols` (3000 obs. of 3 variables), `split_list` (List of 5), `split_matrix` (chr [1:5, 1:2] "Security" "Research" "Construction" "Software" "Fin..."), `starts_with_s` (3000 obs. of 10 variables), `text_data` (5 obs. of 2 variables), and `tidy_data` (5 obs. of 6 variables).
- Files:** Shows a file explorer with folders like `gignore`, `RData`, `Rhistory`, `AI_Impact_on_Jobs_2030.csv`, `analysis.R`, `prac.pdfs`, `README.md`, and `RPROG.Rproj`.
- Code Editor:** Contains R code for importing and manipulating data. The code includes comments and function calls like `library(tidyverse)`, `read.csv()`, `distinct()`, `slice()`, `str_sub()`, and `str_extract()`.

The screenshot shows the RStudio environment with the following components:

- Environment:** Lists objects in the workspace: `final_list` (5 obs. of 3 variables), `job_title` (3000 obs. of 18 variables), `manual_split_df` (5 obs. of 3 variables), `merged_data` (5 obs. of 7 variables), `range_cols` (3000 obs. of 3 variables), `selected_cols` (3000 obs. of 3 variables), `split_list` (List of 5), `split_matrix` (chr [1:5, 1:2] "Security" "Research" "Construction" "Software" "Fin..."), `starts_with_s` (3000 obs. of 10 variables), `text_data` (5 obs. of 2 variables), and `tidy_data` (5 obs. of 6 variables).
- Files:** Shows a file explorer with folders like `gignore`, `RData`, `Rhistory`, `AI_Impact_on_Jobs_2030.csv`, `analysis.R`, `prac.pdfs`, `README.md`, and `RPROG.Rproj`.
- Code Editor:** Contains R code for manipulating and splitting data. The code includes comments and function calls like `str_sub()`, `str_extract()`, `separate()`, and `tidy_data`.

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```
RPROG - RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help

AI Impact on Jobs 2030
Job Title Average Salary Years Experience Education Level AI Exposure Index Tech Growth Factor Automation Probability 2030 Risk C
Showing 1 to 1 of 3,000 entries, 18 total columns

> # R452 - RPROG/
> # Load data from file
> # The code below re-loads the data and creates the same sample used previously
> # for a self-contained demonstration.
> ai_jobs_df <- read.csv("AI_Impact_on_Jobs_2030.csv")

> text_data <- ai_jobs_df %>%
+   distinct(Job_Title, .keep_all = TRUE) %>%
+   slice(1:5) %>%
+   select(Job_Title, Education_Level)

> print("---- Data used for splitting demonstration (First 3 rows) ----")
[1] "---- Data used for splitting demonstration (First 3 rows) ----"
> print(head(text_data, 3))
  Job_Title Education_Level
1 Security Guard Master's
2 Research Scientist PhD
3 Construction Worker High School

> print("-----")

> # 1. RE-LOAD AND SAMPLE DATASET
> # The code below re-loads the data and creates the same sample used previously
> # for a self-contained demonstration.
> ai_jobs_df <- read.csv("AI_Impact_on_Jobs_2030.csv")

> text_data <- ai_jobs_df %>%
+   distinct(Job_Title, .keep_all = TRUE) %>%
+   slice(1:5) %>%
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> print("---- Data used for splitting demonstration (First 3 rows) ----")
[1] "---- Data used for splitting demonstration (First 3 rows) ----"
> print(head(text_data, 3))
  Job_Title Education_Level
1 Security Guard Master's
2 Research Scientist PhD
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> print("-----")

> # 2. USING str_split() (Split String) - The Missing Demonstrations
> # Scenario: Split 'Job_Title' by the space (" ") into multiple components.

> # Method A: Basic Split (Result is a list)
> # This is the raw output, where each split string becomes an element in a list.
> split_list <- str_split(text_data$Job_Title, " ")
> print("---- Method A: Basic Split Output (List format) ----")
[1] "---- Method A: Basic Split Output (List format) ----"
> print(split_list[[1]]) # Viewing the split parts of the first job: "Security Guard"
[1] "Security" "Guard"
> print(split_list[[2]]) # Viewing the split parts of the second job: "Research Scientist"
[1] "Research" "Scientist"

> # Method B: Split Fixed (Returns a matrix)
> # Using simplify = TRUE converts the output to a matrix, which is easy to assign
> # back to new columns using standard R subsetting.
> split_matrix <- str_split(text_data$Job_Title, " ", simplify = TRUE)
> # Assign the split parts to new columns
> manual_split_df <- data.frame(text_data$Job_Title)
> names(manual_split_df) <- "Job_Title"
> manual_split_df$Primary_Role <- split_matrix[, 1] # First column of matrix
```

```
RPROG - RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help

AI Impact on Jobs 2030
Job Title Average Salary Years Experience Education Level AI Exposure Index Tech Growth Factor Automation Probability 2030 Risk C
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> # R452 - RPROG/
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> # for a self-contained demonstration.
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> text_data <- ai_jobs_df %>%
+   distinct(Job_Title, .keep_all = TRUE) %>%
+   slice(1:5) %>%
+   select(Job_Title, Education_Level)

> print("---- Data used for splitting demonstration (First 3 rows) ----")
[1] "---- Data used for splitting demonstration (First 3 rows) ----"
> print(head(text_data, 3))
  Job_Title Education_Level
1 Security Guard Master's
2 Research Scientist PhD
3 Construction Worker High School

> print("-----")

> # 2. USING str_split() (Split String) - The Missing Demonstrations
> # Scenario: Split 'Job_Title' by the space (" ") into multiple components.

> # Method A: Basic Split (Result is a list)
> # This is the raw output, where each split string becomes an element in a list.
> split_list <- str_split(text_data$Job_Title, " ")
> print("---- Method A: Basic Split Output (List format) ----")
[1] "---- Method A: Basic Split Output (List format) ----"
> print(split_list[[1]]) # Viewing the split parts of the first job: "Security Guard"
[1] "Security" "Guard"
> print(split_list[[2]]) # Viewing the split parts of the second job: "Research Scientist"
[1] "Research" "Scientist"

> # Method B: Split Fixed (Returns a matrix)
> # Using simplify = TRUE converts the output to a matrix, which is easy to assign
> # back to new columns using standard R subsetting.
> split_matrix <- str_split(text_data$Job_Title, " ", simplify = TRUE)
> # Assign the split parts to new columns
> manual_split_df <- data.frame(text_data$Job_Title)
> names(manual_split_df) <- "Job_Title"
> manual_split_df$Primary_Role <- split_matrix[, 1] # First column of matrix
```

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The screenshot displays the RStudio interface with the following components:

- Environment:** Lists objects including `final_list` (5 obs. of 3 variables), `job_title` (3000 obs. of 18 variables), `manual_split_df` (5 obs. of 3 variables), `merged_data` (5 obs. of 7 variables), `range_cols` (3000 obs. of 3 variables), `selected_cols` (3000 obs. of 3 variables), `split_list` (List of 5), `split_matrix` (chr [115, 112] "Security" "Research" "Construction" "Software" "Fin..."), `starts_with_s` (3000 obs. of 10 variables), `text_data` (5 obs. of 2 variables), and `tidy_data` (5 obs. of 6 variables).
- Files:** Shows a file explorer with items like `gignore`, `.RData`, `.Rhistory`, `AI_Impact_on_Jobs_2030.csv`, `analysis.R`, `prac.pdfs`, `README.md`, and `RPROG.Rproj`.
- Code Editor:** Contains R code for splitting strings using `str_split()` and `str_split_fixed()`, and creating data frames from the results.

```
R - R432 - RPROG
# Scenario: Split 'Job_Title' by the space (" ") into multiple components.
# Method A: Basic Split (Result is a list)
# This is the raw output, where each split string becomes an element in a list.
split_list <- str_split(text_data$Job_Title, " ")
print(split_list)
# Method B: Split Fixed (Returns a matrix)
# Using simplify = TRUE converts the output to a matrix, which is easy to assign
# back to new columns using standard R subsetting.
split_matrix <- str_split(text_data$Job_Title, " ", simplify = TRUE)
# Assign the split parts to new columns
manual_split_df <- data.frame(text_data$Job_Title,
                             names(manual_split_df) <- "Job_Title")
# Method C: Split Fixed (Returns a matrix)
manual_split_df$Primary_Role <- split_matrix[, 1] # First column of matrix
manual_split_df$Qualifier <- split_matrix[, 2] # Second column of matrix (or empty string if single word title)
print(manual_split_df)
```

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